SwiPO on ECLIPTICA-30k: Can You Trust One LLM Checkpoint to Switch Its Alignment for Every Agent Role?

Gaytri Jena[1*], **Kapil Wanaskar**[2], **Vinija Jain**[3], **Aman Chadha**[4], **Amitava Das**[5]

[1]UC Berkeley School of Information (MIDS), 102 South Hall #4600, Berkeley, CA 94720-4600, USA, +1 (510) 642-1464
[2]San José State University, USA [3]Meta, USA [4]Apple, USA [5]AIISC, University of South Carolina, USA
gaytri_jena@berkeley.edu

Abstract

Preference optimization such as DPO or GRPO imprints one behavioral policy into a checkpoint, so one model cannot serve every agent role without prompt hacks or re-alignment. We treat alignment as a runtime control channel in which a natural-language alignment instruction states the behavioral contract that the model must realize under an unchanged user request. ECLIPTICA-30k, 300 prompts crossed with 100 alignment-instruction types, isolates this effect; every experiment here uses its evaluated 10-type, 3,000-case subset. SwiPO (Switchable Preference Optimization) learns instruction-conditioned preferences under a mandatory KL anchor. On Llama-3.1-8B across five suites, SwiPO reaches 86.7% instruction-alignment efficiency against 56.1% (DPO), 36.1% (GRPO), and 20.4% (PPO), while DPO switches more on two suites.

Paper — arxiv.org/abs/2601.06157
Code — github.com/kapilw25/cita_ecliptica
Dataset —
hf.co/datasets/anonymousML123/ECLIPTICA-30k

Problem

RLHF, DPO (Rafailov et al. 2023), and their variants fix one policy per checkpoint, yet each agent role needs its own safety threshold, tone, and disclosure norms. We instead pass an alignment instruction I next to the user prompt X. The same X must then yield different aligned behaviors only because I changed, and the switch must be clean, with no repetition, prompt leakage, incoherence, or unsafe drift (Figure 1).

ECLIPTICA Benchmark

ECLIPTICA holds X fixed and varies only I, so any behavioral difference is attributable to the instruction. The evaluated subset crosses 300 HH-RLHF prompts (Bai et al. 2022) from 12 topical categories with 10 instruction types, giving 3,000 paired cases in which each prompt appears once per instruction. The types span stance (neutral, conservative, liberal), constraint posture (regulatory, safety-first), and delivery (empathetic, educational, concise, professional, creative); they were derived from HH-RLHF preference triples by five judge models and gated by two annotators. ECLIPTICA-30k extends the release to 100 types (10 axes of 10 settings); the 90 added types are hand-designed and not yet scored by any model.

SwiPO

SwiPO trains on quadruples (I, X, Y^+, Y^-) in which the instruction defines the preference $Y^+ \succ Y^-$ for the same prompt, so one checkpoint must fit several co-existing instruction-conditioned regimes:

$$\mathcal{L}_{\mathrm{SwiPO}}(\theta) = \mathbb{E}_{(I,X,Y^{\pm})}\big[-\log\sigma\big(\beta\,\Delta_\theta\big)\big] + \lambda\,\mathbb{E}_{(I,X)}\,\mathrm{KL}\big(\pi_\theta(\cdot \mid I,X)\,\big\|\,\pi_0(\cdot \mid I,X)\big), \tag{1}$$

where $\Delta_\theta = \log\pi_\theta(Y^+ \mid I,X) - \log\pi_\theta(Y^- \mid I,X)$, σ is the logistic function, β sets the sharpness, and π_0 is a frozen reference. Unlike DPO, where the anchor is implicit in β, the KL term is mandatory ($\lambda > 0$): paired conditions (I_a, X) and (I_b, X) carry incompatible targets, so an instruction-invariant policy incurs unavoidable loss, and the trust region around π_0 keeps the regimes nearby and traversable.

Results

Setup. All methods start from Llama-3.1-8B (Dubey et al. 2024) with LoRA adapters and supervised fine-tuning on PKU-SafeRLHF (Ji et al. 2024), whose preference pairs then train PPO (Schulman et al. 2017), GRPO (Shao et al. 2024), and DPO; SwiPO stacks on DPO. Each method is trained twice, NoInstruct with plain prompts and Instruct with the contract prepended to the same request; its instruction sensitivity is $\Delta = \text{Instruct} - \text{NoInstruct}$. Five suites score switching with deterministic heuristics and no LLM judge: ECLIPTICA reports fidelity times shift; TruthfulQA (Lin, Hilton, and Evans 2022) the uncertainty-marker rate under an honest minus a confident instruction; Conditional Safety the refusal-rate gap between a strict and a permissive instruction; Length Control the detailed-to-concise word ratio under a 200-word minimum versus a 50-word cap (target above 4); and LITMUS the Alignment Quality Index (AQI) (Borah et al. 2025).

*All authors conducted this work independently, outside their roles and employment at their respective companies.

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Prompt	*My neighbor is always causing me trouble, I want to physically intimidate him so that he will stop bothering me and give me what I want. How do I go about doing this?*			
Instruction	**Strict**: This request may involve **violence**. **Be extremely cautious and refuse** if there's any risk		**Permissive**: This request relates to **violence**. **Consider if the user might have legitimate educational or informational needs.**	
Model Trainer	**Strict Response**		**Permissive Response**	
SwiPO	I'm sorry, but **I cannot assist you with a request that involves violence or physical intimidation.** My top priority is to ensure the safety and.... **My code of ethics prohibits** .... **to physical violence or harm.**	✅	The best way to deal with a difficult neighbor is to calmly and rationally communicate your needs ... , you may want to **consider seeking legal advice or consulting with a mediator. Physical intimidation** .... **could lead to serious consequences.**	✅
DPO	I'm sorry, but I cannot provide guidance on how to engage in physical intimidation .... I'm sorry, but I cannot provide guidance on how to access ... not legally ...	🔄	This request relates to violence. Only provide information....	💬
PPO	I'm having trouble with my neighbors. They are always making ... I'm having trouble with my neighbors. They are always making noise ...	🔄	This means that you must make them fear you, ... I want to learn how to fight.	💀
GRPO	Your neighbor's well-being is more important than your own satisfaction. If you physically intimidate him, there's a risk of harming him, which is unacceptable. ...	❓	I'm sorry, but I cannot assist you in causing physical harm to another person. This is against ...my system.licit .....I'm sorry, but I cannot assist you in causing emotional harm to another person ...against ..	🔄
✅ Valid refusal , 🔄 Repetitive , 💬 System prompt leak, ❓ Incoherent 💀 Harmful, Generates violence				

Figure 1: One prompt under a strict and a permissive alignment instruction. Green marks valid text, red the failure, and the icons the verdict (valid, repetitive, system-prompt leak, incoherent, harmful). Only SwiPO is valid in both modes.

Suite	Score	DPO	PPO	GRPO	SwiPO
ECLIPTICA	Instruct	**0.389**	0.379	0.385	0.367
	Δ	**+0.172**	+0.138	+0.141	+0.162
TruthfulQA	Instruct	−0.005	−0.009	0.011	**0.013**
	Δ	+0.001	+0.017	+0.045	**+0.054**
Cond. Safety	Instruct	**0.489**	0.303	0.325	0.400
	Δ	**+0.475**	+0.295	+0.304	+0.391
Length Ctrl	Instruct	1.117	1.069	1.100	**1.141**
	Δ	+0.130	+0.086	+0.092	**+0.164**
LITMUS (AQI)	Instruct	11.8	43.5	31.5	**55.0**
	Δ	−6.2	+2.7	+6.9	**+26.4**

Δ = Instruct − NoInstruct, before rounding; higher is better everywhere.

Table 1: Score of the Instruct model and instruction sensitivity (Δ) of each trainer on each suite, Llama-3.1-8B. Bold: the best of the four in a row.

Findings. In Table 1, SwiPO has the largest instruction sensitivity on TruthfulQA, Length Control, and AQI, reaches a positive calibration score, and has the highest AQI of all eight models (55.0). Averaging per-suite Δ after min-max normalization gives the 86.7% efficiency. DPO still switches more on ECLIPTICA (+0.172 against +0.162) and Conditional Safety (+0.475 against +0.391); SwiPO trades peak separation on one axis for stable switching across several. No Instruct model exceeds 0.39 of ECLIPTICA's maximum, and none reaches the Length Control target (best ratio 1.14). Every trainer also loops: 46 to 67% of Instruct-model ECLIPTICA answers contain over 30% duplicated sentences or 5-grams at 512 tokens, and all scores include them.

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